

WI4014TU – Numerical Analysis for PDE's

Lecture 12

Numerical solution of the wave equation

Scalar wave equation

$$\begin{aligned}\frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} - \Delta u &= f, \quad \mathbf{x} \in \Omega, \quad 0 < t \leq T \\ u(\mathbf{x}, t) &= g(\mathbf{x}, t), \quad \mathbf{x} \in \partial\Omega, \quad 0 < t \leq T \\ u(\mathbf{x}, 0) &= u_0(\mathbf{x}), \quad \mathbf{x} \in \Omega, \quad t = 0 \\ \left. \frac{\partial u(\mathbf{x}, t)}{\partial t} \right|_{t=0} &= v_0(\mathbf{x}), \quad \mathbf{x} \in \Omega, \quad t = 0\end{aligned}$$

Method of lines for wave equation

With FDM/FVM spatial discretization

Discretization in space: $-\Delta u \rightarrow A\mathbf{u}$, $\mathbf{u}(t) \in \mathbb{R}^n$, $A \in \mathbb{R}^{n \times n}$

$$\mathbf{u}'' = -c^2 A\mathbf{u}, \quad \mathbf{u}(0) = \mathbf{u}^0, \quad \mathbf{u}'(0) = \mathbf{w}^0$$

Assume $A = V\Lambda V^{-1}$ exists. Then, $\mathbf{u}(t) = V\mathbf{b}(t)$, and $\mathbf{b}(t) = V^{-1}\mathbf{u}(t)$.

$$\mathbf{b}'' = -c^2 \Lambda \mathbf{b}, \quad \mathbf{b}(0) = V^{-1}\mathbf{u}^0, \quad \mathbf{b}'(0) = V^{-1}\mathbf{w}^0,$$

$$b_j''(t) = -c^2 \lambda_j b_j(t), \quad j = 1, \dots, n$$

$$b_j(t) = a_j e^{ic\sqrt{\lambda_j}t} + d_j e^{-ic\sqrt{\lambda_j}t}$$

$$\mathbf{b}(t) = e^{ic\Lambda^{1/2}t} \mathbf{a} + e^{-ic\Lambda^{1/2}t} \mathbf{d}, \quad \text{use initial conditions}$$

$$\begin{aligned} \mathbf{b}(t) &= \frac{1}{2} \left(e^{ic\Lambda^{1/2}t} + e^{-ic\Lambda^{1/2}t} \right) \mathbf{b}(0) \\ &\quad + \frac{1}{2i} \left(e^{ic\Lambda^{1/2}t} - e^{-ic\Lambda^{1/2}t} \right) \frac{1}{c} \Lambda^{-1/2} \mathbf{b}'(0) \end{aligned}$$

Method of lines solution

$$\mathbf{u}'' = -c^2 A \mathbf{u}, \quad \mathbf{u}(0) = \mathbf{u}^0, \quad \mathbf{u}'(0) = \mathbf{w}^0,$$

$$\mathbf{u}(t) = V \mathbf{b}(t), \quad \mathbf{u}(0) = \mathbf{u}^0, \quad \mathbf{u}'(0) = \mathbf{w}^0$$

$$\begin{aligned} \mathbf{u}(t) &= V \left[\cos \left(c \Lambda^{1/2} t \right) \mathbf{b}(0) + \sin \left(c \Lambda^{1/2} t \right) \frac{1}{c} \Lambda^{-1/2} \mathbf{b}'(0) \right] \\ &= \sum_{j=1}^n \left[b_j(0) \cos \left(ct \sqrt{\lambda_j} \right) + b_j'(0) \frac{1}{c \sqrt{\lambda_j}} \sin \left(ct \sqrt{\lambda_j} \right) \right] \mathbf{v}_j \end{aligned}$$

Conclusion: oscillating behavior, evolution coefficients neither grow nor decay with time (with the diffusion equation we had exponential decay).

Numerical time "integration" of the wave equation

Central-difference time discretization

$$\frac{\partial^2 u}{\partial t^2} = \frac{u^{k+1}(\mathbf{x}) - 2u^k(\mathbf{x}) + u^{k-1}(\mathbf{x})}{h_t^2} + \mathcal{O}(h_t^2)$$

Explicit time-stepping scheme:

$$\frac{u^{k+1}(\mathbf{x}) - 2u^k(\mathbf{x}) + u^{k-1}(\mathbf{x})}{c^2 h_t^2} = \Delta u^k(\mathbf{x}) + f^k(\mathbf{x})$$

$$u^{k+1}(\mathbf{x}) = 2u^k(\mathbf{x}) - u^{k-1}(\mathbf{x}) + (ch_t)^2 \Delta u^k(\mathbf{x}) + (ch_t)^2 f^k(\mathbf{x})$$

Needs both $u^k(\mathbf{x})$ and $u^{k-1}(\mathbf{x})$ to run

Approximation of initial conditions

To start the iterations we need $u^1(\mathbf{x})$ and $u^0(\mathbf{x})$. The initial conditions are

$$u(\mathbf{x}, 0) = u_0(\mathbf{x}), \quad \left. \frac{\partial u(\mathbf{x}, t)}{\partial t} \right|_{t=0} = v_0(\mathbf{x}), \quad \mathbf{x} \in \Omega$$

Hence, $u^0(\mathbf{x}) = u_0(\mathbf{x})$. To approximate $u^1(\mathbf{x})$ consider the Taylor expansion

$$u^1(\mathbf{x}) = u^0(\mathbf{x}) + \left. \frac{\partial u(\mathbf{x}, t)}{\partial t} \right|_{t=0} h_t + \frac{1}{2} \left. \frac{\partial^2 u(\mathbf{x}, t)}{\partial t^2} \right|_{t=0} h_t^2 + \mathcal{O}(h_t^3)$$

Since one-sided FD approximation

$$\frac{u^1(\mathbf{x}) - u^0(\mathbf{x})}{h_t} = \left. \frac{\partial u(\mathbf{x}, t)}{\partial t} \right|_{t=0} + \mathcal{O}(h_t)$$

gives only $\mathcal{O}(h_t)$ approximation of the initial condition, we need to retain one more term in the approximation of $u^1(\mathbf{x})$

$\mathcal{O}(h_t^2)$ approximation of initial conditions

$$\begin{aligned} u^1(\mathbf{x}) &= u^0(\mathbf{x}) + \left. \frac{\partial u(\mathbf{x}, t)}{\partial t} \right|_{t=0} h_t + \frac{1}{2} \left. \frac{\partial^2 u(\mathbf{x}, t)}{\partial t^2} \right|_{t=0} h_t^2 + \mathcal{O}(h_t^3) \\ &= u_0(\mathbf{x}) + h_t v_0(\mathbf{x}) + \frac{(ch_t)^2}{2} [\Delta u_0(\mathbf{x}) + f^0(\mathbf{x})] + \mathcal{O}(h_t^3) \end{aligned}$$

where we have used the wave equation at $t = 0$:

$$\frac{1}{c^2} \left. \frac{\partial^2 u}{\partial t^2} \right|_{t=0} = \Delta u(\mathbf{x}, 0) + f(\mathbf{x}, 0)$$

Hence, given $u_0(\mathbf{x})$, $v_0(\mathbf{x})$, and $f^k(\mathbf{x})$, the time-stepping iterations are:

$$u^0(\mathbf{x}) = u_0(\mathbf{x})$$

$$u^1(\mathbf{x}) = u_0(\mathbf{x}) + h_t v_0(\mathbf{x}) + \frac{(ch_t)^2}{2} [\Delta u_0(\mathbf{x}) + f^0(\mathbf{x})]$$

$$u^{k+1}(\mathbf{x}) = 2u^k(\mathbf{x}) - u^{k-1}(\mathbf{x}) + (ch_t)^2 [\Delta u^k(\mathbf{x}) + f^k(\mathbf{x})]$$

Time-stepping with FDM/FVM Laplacian

After discretization in space $-\Delta u^k(\mathbf{x}) \rightarrow A\mathbf{u}^k$:

$$\mathbf{u}^0, \mathbf{w}^0 \quad - \quad \text{given}$$

$$\mathbf{u}^1 = \mathbf{u}^0 + h_t \mathbf{w}_0 + \frac{(ch_t)^2}{2} [-A\mathbf{u}^0 + \mathbf{f}^0]$$

$$\mathbf{u}^{k+1} = 2\mathbf{u}^k - \mathbf{u}^{k-1} + (ch_t)^2 [-A\mathbf{u}^k + \mathbf{f}^k]$$

Convergence of the three-point time-stepping scheme

- Time-stepping iterations converge to the exact solution on $t \in [0, T]$ as $h_t \rightarrow 0$

Stability of time-stepping for wave equation with FDM Laplacian

Time-stepping with FDM Laplacian

We consider the iterations

$$\mathbf{u}^{k+1} = 2\mathbf{u}^k - \mathbf{u}^{k-1} - (ch_t)^2 A \mathbf{u}^k, \quad \mathbf{u}^k \in \mathbb{R}^n$$

Assuming $A = V\Lambda V^{-1}$ exists, let $\mathbf{u}^k = V\mathbf{b}^k$, same as in the method of lines. Then, we obtain iterations for $\mathbf{b}^k$:

$$\begin{aligned}\mathbf{b}^{k+1} &= 2\mathbf{b}^k - \mathbf{b}^{k-1} - (ch_t)^2 \Lambda \mathbf{b}^k, \quad \mathbf{b}^k \in \mathbb{R}^n \\ b_j^{k+1} &= 2b_j^k - b_j^{k-1} - (ch_t)^2 \lambda_j b_j^k, \quad j = 1, \dots, n\end{aligned}$$

Let $b_j^k = (b_j)^k$, i.e., b_j to the power k . Dividing by $(b_j)^{k-1}$ and solving the quadratic equation for b_j :

$$\begin{aligned}(b_j)^2 - [2 - (ch_t)^2 \lambda_j] b_j + 1 &= 0 \\ b_j &= \frac{1}{2} \left[2 - (ch_t)^2 \lambda_j \pm \sqrt{(2 - (ch_t)^2 \lambda_j)^2 - 4} \right]\end{aligned}$$

Condition on b_j

- In the exact method-of-lines solution $|b_j(t_k)|$ are oscillating in time. There is no exponential decay/growth.
- Hence, in the numerical time stepping we need to have $|b_j| = 1$, so that $|(b_j)^k| = |b_j|^k = 1^k = 1$, since otherwise we get exponential decay/growth of $(b_j)^k$ with k
- For the equation of the form $b_j^2 - \beta b_j + 1 = 0$, the product of the two roots is always equal to 1. To have $|b_j| = 1$ these roots have to be complex conjugates of each other: $b_{j+} = e^{i\theta}$ and $b_{j-} = e^{-i\theta}$
- This is possible if the discriminant is less or equal zero, i.e.

$$0 \leq (ch_t)^2 \lambda_j \leq 4$$

CFL stability condition

Hence the solution will have an oscillating form if:

$$\frac{(ch_t)^2}{4} \lambda_j \leq 1$$

For example, in 1D case:

$$\frac{(ch_t)^2}{4} \frac{4}{h_x^2} \sin^2 \left(\frac{\pi j}{2N} \right) \leq 1$$

$$\frac{(ch_t)^2}{h_x^2} \sin^2 \left(\frac{\pi j}{2N} \right) \leq 1$$

$$\frac{(ch_t)^2}{h_x^2} \leq 1$$

$$0 \leq \frac{ch_t}{h_x} \leq 1$$

This is the *Courant-Friedrichs-Lewy* (CFL) stability condition in 1D, and ch_t/h_x is called the *Courant number*.

Numerical dispersion

- General solution of the numerical wave equation for the mode coefficient b_j^k is:

$$b_j^k = a_j e^{ik\theta} + d_j e^{-ik\theta} = a_j e^{i\frac{\theta}{h_t} t_k} + d_j e^{-i\frac{\theta}{h_t} t_k}$$

- Analytical solution of the continuous wave equation for $b_j(t)$ at time t_k is

$$b_j(t_k) = \tilde{a}_j e^{ic\sqrt{\lambda_j} t_k} + \tilde{d}_j e^{-ic\sqrt{\lambda_j} t_k}$$

- Dividing the quadratic equation for $b_j = e^{i\theta}$ by b_j , we get:

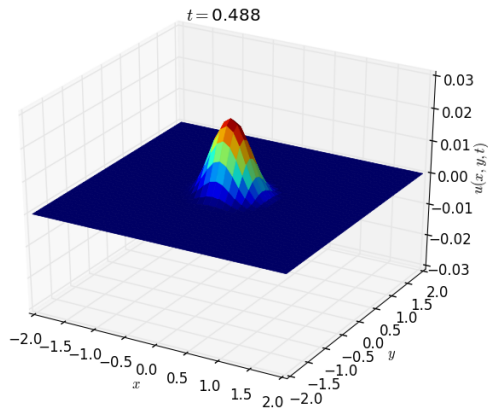
$$b_j - [2 - (ch_t)^2 \lambda_j] + b_j^{-1} = 0 \implies e^{i\theta} + e^{-i\theta} = 2 - (ch_t)^2 \lambda_j$$

$$2 \cos(\theta) = 2 - (ch_t)^2 \lambda_j \implies 4 \sin^2 \left(\frac{\theta}{2} \right) = (ch_t)^2 \lambda_j$$

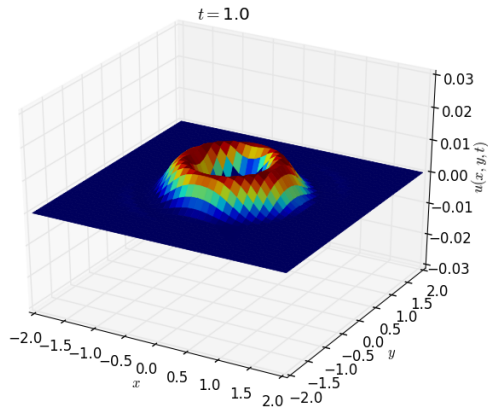
$$\implies \sin \left(\frac{\theta}{2} \right) = \frac{1}{2} ch_t \sqrt{\lambda_j}$$

- Hence, $\frac{\theta}{h_t} \neq c\sqrt{\lambda_j}$, which leads to an erroneous phase velocity for each wave mode.

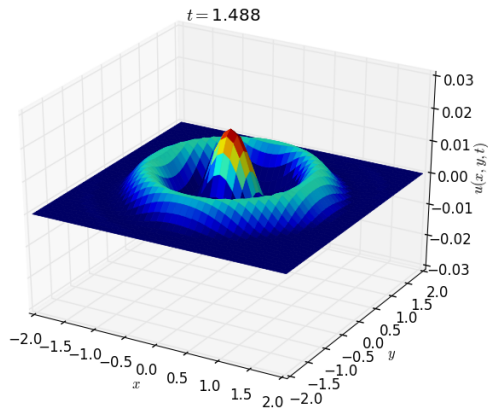
Wave motion: $t = 0.5$



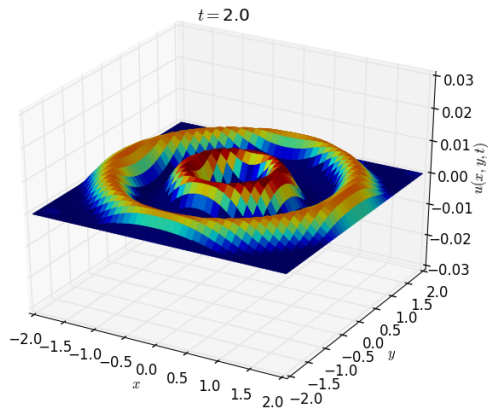
Wave motion: $t = 1.0$



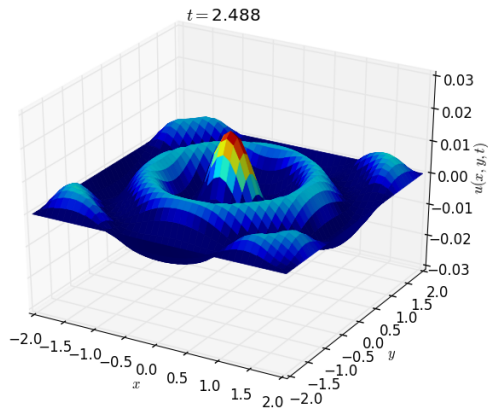
Wave motion: $t = 1.5$



Wave motion: $t = 2.0$



Wave motion: $t = 2.5$



Wave motion: $t = 3.0$

